

30 B

$$\frac{A - X}{A_0 - X} = \left(\frac{1}{2}\right)^{\frac{t}{t_{\frac{1}{2}}}}$$

$$\frac{34 - X}{100 - X} = \left(\frac{1}{2}\right)^{\frac{20}{10}} = \frac{1}{4}$$

$$X = 12 \text{ Bq}$$